

Problem Set 5 — Logic and Proof

VCE Specialist Mathematics 3&4 · Chapter 2 · Weeks 8–10

Due: final week of term. Answers at the back — attempt everything first.

Name: _____

Due: Monday 14 September

Method reminder: written proofs run *forwards*, one justified claim per line. Contrapositive of $P \Rightarrow Q$ is $\neg Q \Rightarrow \neg P$ (equivalent); the converse is not. One counterexample kills a universal claim. Induction: base case, hypothesis, step (which must *use* the hypothesis), conclusion.

Section A — Statements, quantifiers, counterexamples

A1. For the statement “if n is odd, then n^2 is odd”:

a.) Write the converse. Is it true? (2 marks)

b.) Write the contrapositive. Is it true? (2 marks)

A2. Negate each statement (no “it is not the case that”):

a.) For all real x , $x^2 + 1 > 0$.

b.) There exists an integer n such that $n^2 = 2$.

c.) For all $\varepsilon > 0$, there exists $\delta > 0$ such that $\delta < \varepsilon$.

(3 marks)

A3. Find a counterexample to each claim:

a.) For all positive integers n , $n^2 + n + 41$ is prime. (2 marks)

b.) For all integers a, b, c : if a divides bc , then a divides b or a divides c . (2 marks)

Section B — Direct, contrapositive, contradiction

B1. Prove directly: the product of two odd integers is odd.

(3 marks)

B2. Prove by contrapositive: for $x, y \in \mathbb{R}$, if $x + y$ is irrational, then x is irrational or y is irrational. (3 marks)

B3. Prove by contradiction that $\sqrt{3}$ is irrational. (You may use: if $3 \mid n^2$ then $3 \mid n$.)

(4 marks)

Section C — Inequalities, telescoping, induction

C1. Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$ for all real a, b, c . When does equality hold?

(4 marks)

C2. Evaluate $\sum_{k=1}^n \frac{1}{(2k-1)(2k+1)}$, and find its limit as $n \rightarrow \infty$. (4 marks)

C3. Prove by induction: $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$ for all $n \geq 1$. (4 marks)

C4. Prove by induction: 7 divides $8^n - 1$ for all integers $n \geq 1$. (4 marks)

Answers

A1. a.) Converse: if n^2 is odd then n is odd — true (contrapositive of “ n even $\Rightarrow n^2$ even”). b.) Contrapositive: if n^2 is even then n is even — true, equivalent to the original.

A2. a.) There exists a real x with $x^2 + 1 \leq 0$. b.) For all integers n , $n^2 \neq 2$. c.) There exists $\varepsilon > 0$ such that for all $\delta > 0$, $\delta \geq \varepsilon$.

A3. a.) $n = 41$: $41^2 + 41 + 41 = 41 \cdot 43$, composite. (Also $n = 40$: $40^2 + 40 + 41 = 41^2$.) b.) $a = 6$, $b = 4$, $c = 9$: $6 \mid 36$ but $6 \nmid 4$ and $6 \nmid 9$.

B1. $(2m+1)(2n+1) = 2(2mn+m+n)+1$, odd.

B2. Contrapositive: if x and y are both rational, then $x+y$ is rational. $\frac{p}{q} + \frac{r}{s} = \frac{ps+qr}{qs}$, rational. ■

B3. Suppose $\sqrt{3} = p/q$ in lowest terms. Then $p^2 = 3q^2$,

so $3 \mid p^2$, so $3 \mid p$; write $p = 3m$. Then $9m^2 = 3q^2 \Rightarrow q^2 = 3m^2$, so $3 \mid q$ — contradicting lowest terms.

C1. $2(a^2+b^2+c^2) - 2(ab+bc+ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0$. Equality iff $a = b = c$.

C2. $\frac{1}{(2k-1)(2k+1)} = \frac{1}{2} \left(\frac{1}{2k-1} - \frac{1}{2k+1} \right)$; sum = $\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) = \frac{n}{2n+1} \rightarrow \frac{1}{2}$.

C3. Base $n = 1$: $1 \cdot 1! = 1 = 2! - 1$. Step: $\sum^{m+1} = (m+1)! - 1 + (m+1)(m+1)! = (m+1)!(m+2) - 1 = (m+2)! - 1$. ■

C4. Base: $8 - 1 = 7$. Step: $8^{m+1} - 1 = 8(8^m - 1) + 7 = 8(7j) + 7 = 7(8j+1)$. ■